Three Kinds of System Prompt: Comparing What AI Labs Publish About How Their Models Behave

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Abstract

AI labs now publish long documents meant to shape how their models behave. Some are built into the model during training. Some set rules for whose instructions win at runtime. Some are the literal text a product sends with every conversation. People compare these as if they were one kind of thing. I argue they are three, and that mixing them up produces comparisons that don't hold up.

Prior work studies task prompts for benchmarking, attacks that extract system prompts [Yang et al., 2026], and how researchers and policymakers discuss system-level instructions [Neumann et al., 2026]. None of it codes the documents themselves. I read the primary documents from Anthropic, OpenAI, Google DeepMind and xAI, plus system prompts from more than 25 coding-agent products, tiered each source by how far it can be trusted, and coded each one on seven axes. Four things stood out. The labs don't publish the same kind of document. xAI committed to publishing its prompts in response to an incident. Coding agents spend their prompts on tool use and say comparatively little about content safety. And Anthropic's public changelog for its claude.ai prompt, 18 dated revisions over 23 months, shows that at least one lab's "living document" claim can be checked against a dated record.

One person coded the corpus, so I offer this as a first map to be checked and extended, not a settled result.

Keywords: system prompts, LLM alignment, AI governance, prompt engineering, behavioral control, transparency

1 Introduction

Every deployed chatbot runs under instructions written by the company behind it. These sit above whatever the user types, and people usually call them "system prompts." They used to be a few lines. The documents that shape model behavior are now long: Anthropic's constitution for Claude runs to roughly 23,000 words [Anthropic, 2026a], and OpenAI's Model Spec is 128 pages in the revision I analyze [OpenAI, 2025].[1] A whole ecosystem of public repositories also exists just to collect these documents, whether released officially or pulled out through extraction. The largest passed 140,000 GitHub stars at the time of writing [Valbuena, 2026].

For all that attention, few people have compared the documents themselves. Three nearby literatures exist. The first is prompt engineering, which builds taxonomies of *task* prompts, the instructions a user or

[1] I analyze the 2025-09-12 revision throughout. OpenAI has kept revising the Model Spec since, with updates through at least March 2026. As with every artifact in this study, I report a dated snapshot, not a live copy; Section 4 explains how I handle snapshots.

developer writes to get a particular output, mostly for benchmarking and comparing techniques [Karmaker Santu and Feng, 2023, Sahoo et al., 2024]. The second is AI security, which treats system prompts as a *target*: how adversarial queries can pull them out, and how to prevent that [Yang et al., 2026, Zhuang et al., 2025]. The third, and the closest to this paper, is governance studies. Neumann et al. [2026] review how researchers and policymakers *talk about* system-level instructions as a way to govern AI, but they do not code the primary documents and compare them across labs. So the question I ask is still open: across labs and product types, what do these documents contain, how are they built, and what does the variation say about how each company approaches governance?

Three reasons make this worth doing. For governance, these documents are some of the few concrete things the public can inspect to see how a lab turns its stated values into instructions. They are more specific than a blog post and more current than a research paper. For NLP research, using governance documents as training or evaluation data (some labs already do this internally) needs a shared vocabulary for what they contain, and the public literature doesn't have one yet. For engineers, developers building on these models often mix up different kinds of document, for example by setting a leaked *deployed* prompt from one product next to an official *training-time* document from another. Those two can't be compared directly.

Contributions

1. I propose a three-part taxonomy that separates training-time constitutions, runtime authority specifications, and deployed inference-time system prompts, and I show that comparisons across the three go wrong.

2. I build a source-tiered dataset covering four labs and a corpus of coding-agent products, coded on seven axes: identity framing, safety and refusal instructions, tool-use governance, tone and persona control, output format, injection countermeasures, and recency handling.

3. I report cross-lab findings: labs differ in whether they publish governance documents at all and in which kind; one lab's transparency was a reaction to an incident; consumer chat and coding-agent products stress different things; and one lab's public, dated changelog (18 revisions in 23 months) lets me measure publication cadence directly, instead of taking its "living document" description on trust.

4. I say what I would study next, and what would make these results stronger.

2 Related Work

2.1 Task-Prompt Taxonomies

Most prompt-taxonomy work is about structuring task instructions so models score better on benchmarks. The TELeR taxonomy [Karmaker Santu and Feng, 2023] sorts prompts by turn structure, expression style, level of detail and role, so that benchmark results can be reported consistently. Broader surveys [Sahoo et al., 2024] catalog techniques such as chain-of-thought, few-shot examples and retrieval augmentation. A related line catalogs prompt *defects*, the ways task prompts fail, drawing on software-engineering and NLP venues [Tian et al., 2025]. All of it treats the prompt as something a user or developer writes to get a job done. None of it covers the instruction layer written by the provider, which is what I study.

2.2 Prompt Security and Extraction

A large, active literature treats system prompts as secrets worth protecting. Early work separated prompt injection from prompt leaking as two kinds of attack [Perez and Ribeiro, 2022]. Later studies measured

leakage across many deployed applications and found rates above 80% in some large-scale settings [Yang et al., 2026], and proposed defenses such as proxy substitution [Zhuang et al., 2025] and encoding-based hardening [Sahu et al., 2026]. These papers study the *event* of extraction and how well a prompt holds up, not what the prompt says. My work complements them. Security research asks whether a prompt can be extracted. I ask what it says once it is available through legitimate or well-established channels, and what patterns show up across many of them.

2.3 AI Governance Transparency

A smaller literature, along with some industry practice, deals with transparency in AI governance more broadly: model cards, system cards, published behavioral specifications. OpenAI's Model Spec presents itself as a transparency artifact that governs the "chain of command" between the provider, operators (the authors of system prompts), and users [OpenAI, 2025]. Anthropic's constitution describes itself as written mainly *to* the model, favoring precision over public readability, with public release as a secondary benefit [Anthropic, 2026a].

The closest prior work is a contemporaneous study by Neumann et al. [2026], published at FAccT 2026. They treat system-level instructions as a governance object. Their method is a PRISMA-style systematic review of 287 papers, which yields an eight-category typology of the *goals* researchers attribute to such instructions (alignment, stability and auditability, for example), plus a policy analysis of two cases: US Executive Order 14319 and the EU General-Purpose AI Code of Practice. Their main finding is that governance frameworks often assume system-level instructions give stable, predictable control, and the technical literature does not consistently support that. That is a claim about the literature and the policy discussion *around* these documents. They organize their typology by the *stakeholder hierarchy* through which instructions are issued, a "prompt stack" of foundation-model provider, developer or deployer, and end user.

My study differs in scope and method, and does not contradict theirs. I don't survey what researchers or policymakers claim. I code the content of the primary documents and compare them, organized by *artifact type* (training-time, runtime-authority, deployed), which cuts across their stakeholder axis. From the sections and reference list available to me when I wrote this, their corpus uses Anthropic's published deployed system prompts but does not seem to treat Anthropic's training-time constitution as a separate kind of artifact. That fits my taxonomy, where the constitution is a different object from the runtime and deployed layers that their framing describes best. I see the two papers as complementary. Theirs shows, at the level of research and policy, that the premise of prompt-based governance is more contested than people assume. Mine gives a primary-source account of what the documents behind that premise contain.

3 A Taxonomy of Behavioral-Control Artifacts

Informal comparisons of "system prompts" across labs mix up three different kinds of artifact. I propose the following taxonomy, based on the sources I examined.

Training-time constitutions. Documents used during training to shape the model's weights, usually through constitutional-AI-style methods where the document generates synthetic training signal such as example interactions, response rankings and scenario guidance. Anthropic's constitution is the clearest public case. Claude is its main audience, it is not injected at inference time, and it avoids prescribing rules in favor of reasoning the model is meant to absorb [Anthropic, 2026a].

Runtime authority specifications. Documents that say how a deployed model should weigh instructions from different sources at inference time. They are not a single prompt themselves. OpenAI's Model Spec is the clearest case. It defines a ranked "principal hierarchy" (provider, operator, user) and

explains how conflicts between levels should resolve, but it is not the text sent in any one conversation [OpenAI, 2025].

Deployed inference-time system prompts. The actual text placed at the start of a conversation for a given product at a given time. This is the artifact most often leaked or extracted and, in xAI's case [xAI, 2026], published in raw form on purpose. It is the most short-lived of the three, changing with product updates, and the most varied, since one lab can run dozens of different deployed prompts across its products.

The taxonomy is more than bookkeeping. Section 5 shows that mixing the types produces comparisons that don't hold up, and that which type a lab picks for a given constraint says something about how that lab thinks about governance.

4 Methodology

4.1 Source Tiering

The corpus mixes documents of different provenance, so each entry carries a credibility tier, which I report openly instead of treating every source as equally authoritative. These tiers measure how far a source can be trusted. They are separate from the three artifact types in Section 3.

- **Tier 1 (Official/primary):** Content published directly by the lab that wrote it (for example Anthropic's constitution, OpenAI's Model Spec, xAI's raw prompt repository).
- **Tier 2 (Reconstructed, high-consensus):** Content that shows up consistently across independent extraction attempts and lives in long-running, widely cited public repositories (for example the coding-agent prompt collection studied in Section 5.5).
- **Tier 3 (Single-source leaks):** Lower-confidence material, used only to note that a pattern may exist and never as a primary data point.

4.2 Coding Scheme

Each entry is coded on seven axes: identity framing, safety and refusal instructions, tool-use governance, tone and persona control, output format constraints, injection and jailbreak countermeasures, and recency and knowledge-cutoff handling. Document length and revision history are not coded as axes. Where a source gives them (the Model Spec's page count, the constitution's word count, Anthropic's changelog), they sit in the notes column of the dataset. I derived the scheme inductively from a first pass over the corpus instead of fixing it in advance, which is standard practice in qualitative coding of emerging categories [Tian et al., 2025].

4.3 Copyright and Reproduction Policy

I don't reproduce system prompt text beyond short quoted fragments where the exact wording matters for the analysis. Everything else is paraphrase and structural coding, because I treat these documents as evidence, not as text to republish.

4.4 Scope and Corpus

The corpus (Table 1) covers four labs at the level of official or near-official governance documents, plus more than 25 coding-agent products whose prompts come from a single long-running, widely cited Tier-2 collection [Valbuena, 2026]. I use the coding-agent set to describe patterns by product category, not to make claims about any single product.

Lab	Artifact	Tier	Type (per Sec. 3)
Anthropic	Constitution	1	Training-time
Anthropic	`claude.ai`/mobile system-prompt changelog	1	Deployed (versioned, 18 dated revisions, Jul. 2024–Jun. 2026)
OpenAI	Model Spec	1	Runtime authority spec
xAI	`grok-prompts` repo	1	Deployed (multiple, raw)
Google DeepMind	Prompting guides	3	None published (developer-facing only)
25+ coding-agent products	Aggregated repo	2	Deployed (per-product)

Table 1: Corpus overview by lab or product category, credibility tier, and artifact type. Anthropic is the only lab with both a training-time document and a versioned, dated record of its deployed prompt.

5 Findings

5.1 Labs Publish Different Kinds of Document, Not Just Different Content

The most basic finding is that labs don't agree on which kind of document should do the governing. Anthropic's main public artifact is a training-time document that is never injected at runtime. OpenAI's is a runtime framework for deciding whose instructions win, which is not itself a prompt. xAI's is the deployed prompts, published raw, with no public training-time or arbitration document alongside. Google DeepMind has published nothing in any of the three categories about its own constraints on Gemini. Its public documentation is prompting guidance for outside developers, not a statement about how Gemini is governed. So a comparison of "Anthropic's system prompt versus OpenAI's versus Google's" doesn't line up four instances of one thing. At best it lines up two comparable ones and two gaps.

5.2 Same Structure, Different Reasons

Even with different artifact types, Anthropic and OpenAI use a similar device: an explicit ranking of whose instructions come first. OpenAI's principal hierarchy ranks provider, operator and user instructions, and higher levels win when they conflict [OpenAI, 2025]. Anthropic's constitution sets a priority order of safety, ethics, compliance and helpfulness. They justify it very differently. OpenAI talks about steerability and developer control, and compares the structure to function definitions and arguments in a programming language. Anthropic presents its version as reasoning the model should absorb, not rules to apply mechanically, and says it prefers explanation over rule lists because rule lists don't generalize to new situations. The structure looks alike but the philosophy is not, and an analysis that only asked "does the document specify a hierarchy?" would miss the difference.

5.3 Transparency Can Be a Reaction

Press coverage often describes xAI's publication of Grok's raw system prompts as a transparency move on par with Anthropic's. Where it came from is different. xAI announced the commitment in direct response to a May 2025 incident, in which an unauthorized internal change led the Grok bot on X to give off-topic, ideologically directed replies [xAI, 2025]. It framed publication as a way to rebuild trust and paired it with more internal review controls. That is not the principled, forward-looking stance behind Anthropic's and OpenAI's releases. So "publishes its system prompts" covers at least two things: transparency as a value, and transparency as incident response. They may deserve different levels of trust as a picture of how the model normally behaves. One public issue thread on xAI's repository points the same way. It describes a prompt the model reportedly shared in conversation that was never found in the repository. I don't try to settle what that means. I flag it because it shows that Tier-1 status does not guarantee completeness.

5.4 Content Differs, Not Just Structure

Labs also differ in what their constraints actually say, and the differences match how each company positions its product, not just how it writes. xAI's published safety-prefix file for its consumer model says there are no added content restrictions on fictional adult sexual content involving dark or violent themes unless specified elsewhere. It also tells the model to treat users as adults and not to moralize about "edgy" requests, and it marks these instructions as the highest priority [xAI, 2026]. That is a more permissive default for mature fiction than anything publicly documented by the other labs in this corpus. The same material keeps a narrower, explicitly named restriction on content involving minors. Broad permissiveness with one hard, named exception looks like a deliberate product choice, not an oversight, and the primary source says it plainly enough that I don't have to infer it.

5.5 Product Category Predicts What a Prompt Emphasizes

Coding-agent products (Cursor, Claude Code, GitHub Copilot, Devin and more than twenty others, all from one widely cited Tier-2 collection) look different from consumer chat assistants. Their prompts put heavy weight on governing tools: the order of tool calls, confirmation before destructive file operations, diff-format conventions. They say comparatively little about general content safety. I read this as fitting the three-artifact taxonomy in Section 3. If the base model already carries general safety behavior from a training-time layer, the product's deployed prompt has a narrower job: governing what is specific to that product's task. This fits a layered split of governance work across the three artifact types. It does not prove one.

5.6 Publication Cadence Shows That "Living Document" Can Be Checked

The Anthropic constitution and the OpenAI Model Spec both describe themselves as living documents that will be revised. Anthropic is the only lab in this corpus where that claim can be checked against a public, dated record instead of taken on the publisher's word. Its changelog for the deployed `claude.ai` and mobile system prompt lists 18 dated revisions over about 23 months (July 2024 to June 2026). That is a mean gap of roughly 41 days, under six weeks, between documented changes [Anthropic, 2026b]. The changelog also says these updates do not carry over to the API. That confirms, without any inference from me, that the deployed prompt belongs to a particular product surface and is not one canonical text shared across everything a lab ships, which matches the taxonomy in Section 3. No comparable public record exists for OpenAI, Google DeepMind or xAI at the time of writing, so I can't compare update cadence across labs. I report this as a single case. It shows that at least one lab's "living document" claim can be verified independently, and it shows what verifiable transparency could look like for the others.

6 Discussion

Two things follow from the findings. First, "comparing system prompts" only makes sense once you say which artifact type you mean. Any comparison that crosses types, such as a leaked deployed prompt against an official training-time document, should say so instead of presenting itself as like for like. Second, the differences between labs are not just style. Which artifact type a lab chooses, whether it publishes anything, and the circumstances of publication (planned, or forced by an incident) all tell you something about its approach to governance. Public discussion tends to flatten this into one question, "does lab X publish its prompts?", and to treat the answer as a yes-or-no marker of trustworthiness. It isn't one.

6.1 Implications for Practitioners

Developers who build on several providers' APIs, including agent systems that call more than one model, should know that a product's "safety behavior" is not encoded at the same layer for every provider. A prompt that works under one product's conventions, such as the typically light safety framing of a coding agent, shouldn't be assumed to carry over to a different deployment without checking which layer is supposed to carry that responsibility.

6.2 Implications for Researchers

I suggest treating these documents as a legitimate, citable object of NLP and AI-governance research. The credibility tiers used here are one candidate norm for handling the mix of official and reconstructed material that this area will keep having until more labs publish everything proactively.

7 Limitations

Several limits affect how much weight the findings can bear. First, the corpus is not exhaustive. It covers four labs in depth and one aggregated coding-agent source, not the full range of deployed LLM products. Second, Tier-2 and Tier-3 material carries less evidentiary weight than Tier-1 by construction. Conclusions that rest mostly on Tier-2 sources, especially the coding-agent finding in Section 5.5, are suggestive until someone replicates them against official sources, if labs publish any. Third, most of the artifacts are one or a few snapshots, not time series. Publishers describe the constitution and the Model Spec as living documents, but only Anthropic's deployed-prompt changelog (Section 5.6) gives public, dated evidence of revision in practice. Findings on emphasis or structure for OpenAI, Google DeepMind and xAI may not hold at a later snapshot, and I have no public basis for guessing how fast they would change. Fourth, I derived the coding scheme from the corpus and applied it alone. I did not measure inter-rater reliability, and a second coder would make future versions stronger. Fifth, the interpretation in Section 5.5, that base-model training absorbs general safety behavior while deployed prompts handle task-specific governance, fits the pattern I saw but was not tested against a case that could prove it wrong, such as a coding agent built on a base model with no comparable training-time safety layer. It is the simplest explanation the public evidence allows, not a demonstrated cause.

8 Conclusion and Future Work

The documents that govern large language models have grown from informal instructions into long public artifacts, yet few studies compare their primary text directly, as opposed to reviewing how they are discussed in the literature and in policy [Neumann et al., 2026]. I introduced a three-part taxonomy that separates

training-time constitutions, runtime authority specifications and deployed system prompts, and showed that the separation clears up comparisons across labs that otherwise don't line up. Using a source-tiered method across four labs and a broad set of coding-agent products, I found that labs differ in which kind of artifact they publish at all, that transparency can be a reaction to an incident, that differences in content track product positioning and not only writing style, that product category predicts what a deployed prompt emphasizes, and that at least one lab's claim to maintain a living governance document can be checked against a public, dated record.

Future work should add more labs and product types as official material appears, measure inter-rater reliability on the coding scheme, and follow these artifacts over time, since their publishers call them living documents. That would let the field study how behavioral-control design changes, not only how labs differ at one moment.

Use of AI Tools

I used Claude (Anthropic) to help gather sources and to draft and edit text. I take responsibility for the final text and any errors in it.

References

Anthropic. *Claude's Constitution*. January 2026. https://www.anthropic.com/constitution

Anthropic. *System Prompts* [release-notes documentation, versioned changelog of the deployed claude.ai/mobile system prompt, dated entries from July 2024 through June 2026]. Accessed July 2026. https://docs.claude.com/en/release-notes/system-prompts

A. Neumann, H. Sargeant, and J. Singh. Prompt Governance? On Governing Technologies Governed by Natural Language. *arXiv:2606.07539*, 2026. Also published in *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency* (FAccT '26), pp. 1–44. https://doi.org/10.1145/3805689.3806763

OpenAI. *The OpenAI Model Spec* (2025-09-12 revision). September 2025. https://model-spec.openai.com/2025-09-12.html

L. Valbuena (x1xhlol). *system-prompts-and-models-of-ai-tools* [GitHub repository, aggregating reconstructed system prompts for 25+ coding-agent products; 141k+ stars, 501 commits at time of access]. Accessed July 2026. https://github.com/x1xhlol/system-prompts-and-models-of-ai-tools

xAI. *grok-prompts* [GitHub repository, official raw system prompts for Grok across chat, X-bot, and API deployment surfaces]. Accessed July 2026. https://github.com/xai-org/grok-prompts

xAI. Statement on unauthorized modification to the Grok response bot on X. Posted to X, May 15, 2025. https://x.com/xai/status/1923183620606619649

S. K. Karmaker Santu and D. Feng. TELeR: A General Taxonomy of LLM Prompts for Benchmarking Complex Tasks. *arXiv:2305.11430*. Accepted to Findings of EMNLP 2023.

P. Sahoo, A. K. Singh, S. Saha, V. Jain, S. Mondal, and A. Chadha. A Systematic Survey of Prompt Engineering in Large Language Models: Techniques and Applications. *arXiv:2402.07927*, 2024.

H. Tian, C. Wang, B. Yang, L. Zhang, and Y. Liu. A Taxonomy of Prompt Defects in LLM Systems. *arXiv:2509.14404*, 2025.

F. Perez and I. Ribeiro. Ignore Previous Prompt: Attack Techniques For Language Models. *arXiv:2211.09527*, 2022. Presented at the NeurIPS 2022 ML Safety Workshop.

Y. Yang, C. Fu, T. Zhang, R. Zeng, Q. Li, T. Du, Z. Wang, S. Ji, and W. Chen. Understanding and Mitigating Prompt Leaking Attacks in Real-World LLM-Based Applications. *arXiv:2606.18673*, 2026.

Z. Zhuang, M.-I. Nicolae, H.-P. Wang, and M. Fritz. ProxyPrompt: Securing System Prompts against Prompt Extraction Attacks. *arXiv:2505.11459*, 2025.

A. Sahu, D. Samanta, and R. Soosahabi. Automated Framework to Evaluate and Harden LLM System Instructions against Encoding Attacks. *arXiv:2604.01039*, 2026.